**Homework 10**
**Due**: Saturday, 3rd August 2024 at 1:00 PM

**Instructions.** Provide your answers **and an explanation** to the following questions and *submit as a single PDF on Gradescope* by the due date and time listed above. *Assign pages accordingly when you submit. It is recommended that you type as much as you can to ensure that your solution is legible. Illegible solutions will not receive credit.*

**Note on Academic Dishonesty.** Although you are allowed to discuss these problems with other people, your work must be entirely your own. It is considered academic dishonesty to read anyone else's solution to any of these problems or to share your solution with another student or to look for solutions to these questions online. See the syllabus for more information on academic dishonesty. If it is determined that you have violated any of these guidelines, you will be reported to the Dean of Students and the recommended sanction will be a failing grade in the course.

**Grading.** Some of these questions may be graded for accuracy and some for completion.

**Question 1.**

How many weighings of a balance scale are needed to find a lighter counterfeit coin among four coins? Describe an algorithm to find the lighter coin using this number of weighings.

**Question 2.**

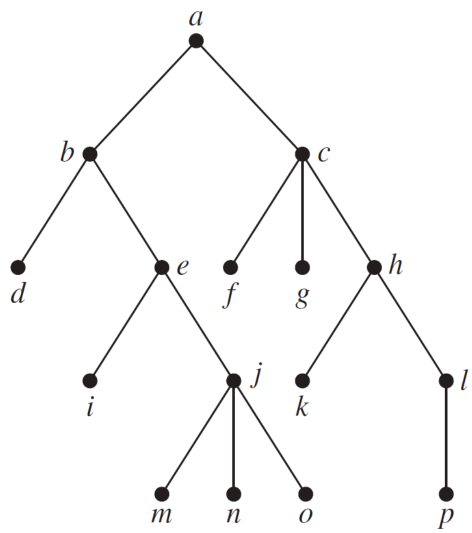
Use Huffman coding to encode these symbols with given frequencies: A: 0.10, B: 0.25, C: 0.05, D: 0.15, E: 0.30, F: 0.07, G: 0.08. What is the average number of bits required to encode a symbol?

**Question 3.**

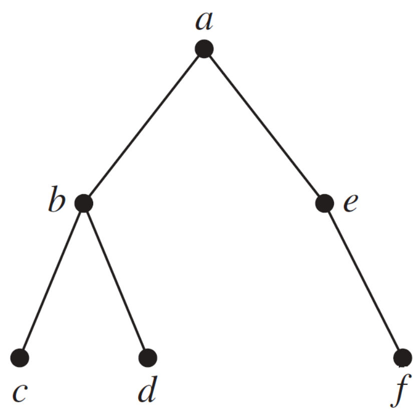
Draw a game tree for nim if the starting position consists of three piles with one, two, and three stones, respectively. When drawing the tree, represent by the same vertex the symmetric positions that result from the same move. Find the value of each vertex of the game tree.Who wins the game if both players follow an optimal strategy?

**Question 4.**

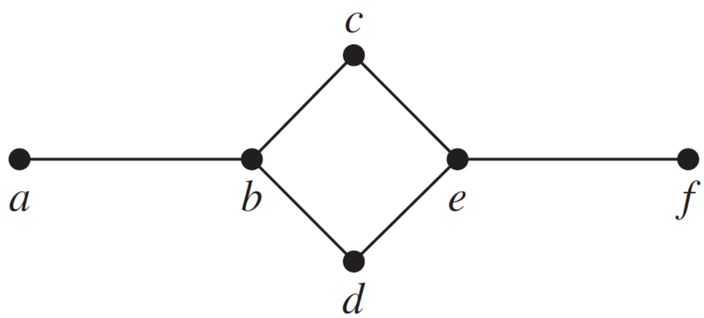
Provide the preorder traversal of the vertices of the following tree. Use the recursive algorithm and show the steps of the algorithm.



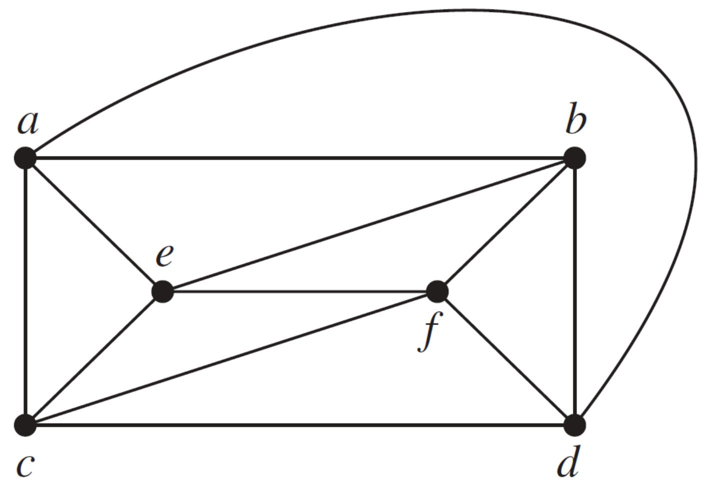
**Question 5.**
Provide the preorder traversal of the vertices of the following tree. Use an iterative process that goes through the boundary of the edges; Show the steps of the process.



**Question 6.**
Draw all the spanning trees of the given simple graph.

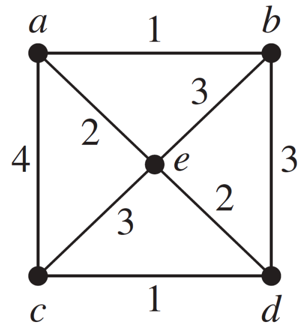


**Question 7.**
Use backtracking to try to find a coloring of the graphs shown below using three colors.



**Question 8.**

Use Prim's algorithm to find a minimum spanning tree for the given weighted graph.



**Extra credit**

Let T be a rooted tree. We assume for simplicity that each internal node has exactly two children. The **lowest common ancestor** of any two nodes u and v is the node that is an ancestor of both u and v, and that is farthest away from the root of T . For any node u of T , let size(u) denote the number of leaves in the subtree of u, and let $l(u) = \lfloor \log size(u) \rfloor$. We will use the values l(u) to partition the tree T into a collection of pairwise disjoint paths. For any node u of T , let $P_u$ be the subgraph of T consisting of all nodes v for which (i) l(v) = l(u), and (ii) l(w) = l(u) for all nodes w on the path in T between u and v. The root of $P_u$ is the group parent of u. **Prove that** if two distinct nodes u and v have the same group parent, then either u is an ancestor of v, or v is an ancestor of u.

Notice that, if you answer the extra credit question correctly, the grade of the extra credit will not show up on Gradescope. But we will add that grade in D2L at the end of the semester.